

Time-calibrated Desmognathus phylogeny with audited bootstrap trait distributions

Size panel: all 21 species, with 50 manually vetted largest cells and corresponding nuclei each. Genome panel: 20 image-quality-matched species calibrated as species/fuscus IOD × 16.36 pg; D. ochrophaeus has no genome estimate because image-quality overlap was limited. No phylogenetic fills.

Descriptive species-level Spearman ρ: genome–nucleus 0.45; genome–cell 0.55; nucleus–cell 0.56. These correlations are not phylogenetically corrected.

Time-calibrated phylogeny

Genome size

Nucleus area

Cell area

Bootstrap estimator distribution    Current point estimate    Across-species median

